go-On Faculty Timetable Desktop Blank Page Fix Report

Defect Investigation, Layout Protocol Root Cause Analysis & Multi-Viewport Verification

Target System: go-On AI Face Attendance	**Date:** October 19, 2026
Affected Route: Faculty Timetable Tab (Index 3)	**Environment:** Flutter Web (Port 3000) & Android
Root Cause: Scaffold BottomNav Unbounded Expansion	**Resolution Status: RESOLVED & VERIFIED**

1. Executive Summary

A critical rendering defect was identified where the **Faculty Timetable page appeared completely blank on laptop and desktop screens** (width >= 900px) when accessed at http://localhost:3000. While the AppBar header ('Faculty Timetable') and the floating bottom navigation bar rendered properly, the middle content area containing schedule, day tabs, and periods was invisible. This report documents the diagnostic tracing, root cause identification within Flutter's Scaffold layout engine, the clean architectural fix applied, and multi-viewport automated verification across six screen configurations.

2. Defect Reproduction & Architectural Investigation

Tracing rendering flow through mobile/lib/faculty/screens/faculty_dashboard_screen.dart and constructing a headless widget test harness at 1680x900 viewport reproduced the exact symptoms observed in the user screenshot:

* **Visual Symptom:** Top AppBar and Bottom Navigation rendered, but body content was invisible.

* **Harness Diagnostic Output:**
  Body Align constraints: BoxConstraints(0.0 <= w <= 1680.0, h = 0.0), size: Size(1680.0, 0.0)
  _RawIndexedStack size: Size(1080.0, 0.0)
  ListView rect: Rect.fromLTRB(300.0, 250.0, 1380.0, 250.0) (Height: **0.0 px**).

The diagnostic proved conclusively that this was not a backend data failure or state management issue; the timetable data was loaded in memory, but the Flutter layout engine collapsed the entire body height to zero pixels.

3. Root Cause Analysis: Flutter Scaffold Layout Engine

The defect originated in the desktop-responsive branch introduced in FacultyDashboardScreen._buildBody and Scaffold.bottomNavigationBar:

```
// PREVIOUS DEFECTIVE CODE IN faculty_dashboard_screen.dart
bottomNavigationBar: isDesktopOrLaptop
    ? Align(
        alignment: Alignment.bottomCenter,
        child: ConstrainedBox(
          constraints: const BoxConstraints(maxWidth: 580.0),
          child: bottomNavWidget,
        ),
      )
    : bottomNavWidget,
```

Why this caused height collapse:
1. In Flutter's ScaffoldLayout, the bottomNavigationBar slot is provided with bounded constraints: BoxConstraints(maxWidth: screenWidth, maxHeight: screenHeight).
2. An Align widget without an explicit heightFactor defaults to shrinkWrapHeight = false. In bounded height, it expands to fill the entire available vertical space (clamped to maxHeight = 900.0 px).
3. The Scaffold layout engine subtracts bottom navigation height from total screen height to allocate space for body: availableBodyHeight = 900.0 - 56.0 (AppBar) - 900.0 (bottomNav) = 0.0 px.
4. As a result, the body received a rigid height constraint of h = 0.0, compressing the IndexedStack and child FacultyTimetableScreen to 0 height, rendering the screen completely empty.

4. Implementation of the Architectural Fix

To resolve layout collapse cleanly while preserving centered pill navigation, the following changes were applied to mobile/lib/faculty/screens/faculty_dashboard_screen.dart:

```
// RESOLVED CODE IN faculty_dashboard_screen.dart
// 1. Constrain bottom navigation to intrinsic height using heightFactor: 1.0
bottomNavigationBar: isDesktopOrLaptop
    ? Align(
        alignment: Alignment.bottomCenter,
        heightFactor: 1.0,
        child: SizedBox(width: 580.0, child: bottomNavWidget),
      )
    : bottomNavWidget,

// 2. Center body content cleanly with maximum readable width (1080px)
body: isDesktopOrLaptop
    ? Center(
        child: ConstrainedBox(
          constraints: const BoxConstraints(maxWidth: 1080.0),
          child: IndexedStack(index: _currentIndex, children: screens),
        ),
      )
    : IndexedStack(index: _currentIndex, children: screens),
```

Engineering Benefits:
* heightFactor: 1.0 forces Align to take exactly the vertical height of its child pill (~70px), allowing Scaffold to allocate the remaining 774px to the body.
* Center combined with BoxConstraints(maxWidth: 1080.0) ensures proper centering on ultra-wide screens without infinite vertical layout fighting.
* Zero regression on mobile devices (width < 900px), where the unconstrained mobile layout executes unmodified.

5. Multi-Viewport Automated Verification Matrix

An automated test suite (mobile/test/desktop_timetable_test.dart) was executed across all key viewport form factors. All tests passed with full layout rendering and confirmed non-zero height.

Viewport Configuration	Resolution	Category	ListView Rect / Height	Result
Desktop Widescreen	1680 x 900	Desktop	Height: 579.6 px	**PASSED**
Standard Laptop	1366 x 768	Laptop	Height: 447.6 px	**PASSED**
Tablet Landscape / Portrait	768 x 1024	Tablet	Height: 703.6 px	**PASSED**
Samsung Galaxy S24 / Modern	390 x 844	Mobile	Height: 523.6 px	**PASSED**
Compact Mobile	360 x 800	Mobile	Height: 479.6 px	**PASSED**
Full HD Ultra-Wide	1920 x 1080	Large Desktop	Height: 759.6 px	**PASSED**

6. Code Quality, Lints & Build Status

* **Static Analysis:** flutter analyze passed with **zero errors and zero warnings**.
* **Data Integrity:** No mock data hardcoded; production API contracts and model bindings preserved intact.
* **Timezone Consistency:** Strict Asia/Kolkata (IST) 12-hour AM/PM formatting maintained across all tabs.
* **Web Build:** Production Flutter Web bundle successfully generated into mobile/build/web.
* **Git Policy:** Strict compliance maintained with zero commits and zero pushes.

Verification Engineer: Antigravity Autonomous Agent	**Build Target:** Flutter Web & Mobile
Status: PRODUCTION READY - ALL CHECKS PASSED	**Date:** October 19, 2026